CHAPTER 9

AMINES

CHAPTER ESSENCE AT A GLANCE:
Structure and basicity of aliphatic and aromatic amines, Gabriel phthalimide synthesis, Hoffmann bromamide reaction, and diazonium salt reactions.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Amine Basicity & Aqueous Influences:** Aliphatic amines are more basic than ammonia due to inductive electron release by alkyl groups, with aqueous basicity governed by an interplay of inductive, steric, and hydration effects; aromatic amines are substantially weaker bases due to resonance delocalization of the nitrogen lone pair.

• **Gabriel Phthalimide & Hoffmann Synthesis:** Gabriel synthesis provides pure primary aliphatic amines without secondary or tertiary contamination, while Hoffmann bromamide degradation shortens the carbon chain by one atom during conversion of primary amides.

• **Carbylamine & Hinsberg Distinctions:** Primary amines form foul-smelling isocyanides in the carbylamine test; Hinsberg's benzenesulphonyl chloride test distinguishes primary, secondary, and tertiary amines by alkali solubility of sulfonamide derivatives.

• **Diazotisation & Synthetic Versatility:** Primary aromatic amines react with nitrous acid at ice-cold temperatures to yield stable arenediazonium salts, which serve as versatile synthetic intermediates in Sandmeyer, Gattermann, and reduction reactions.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$K_b = \frac{[\mathrm{RNH_3^+}][\mathrm{OH^-}]}{[\mathrm{RNH_2}]} \rightarrow \mathrm{p}K_b = -\log_{10} K_b \quad (\text{Basicity Order})$$

Amine basicity equilibrium constant and aqueous pK b relation

$$\text{Basicity in water: } (\mathrm{CH_3})_2\mathrm{NH} > \mathrm{CH_3NH_2} > (\mathrm{CH_3})_3\mathrm{N} > \mathrm{NH_3} \quad (2^\circ > 1^\circ > 3^\circ)$$

Aqueous basicity order for methylamines (Inductive, solvation, and steric effects)

$$\mathrm{R\text{-}CONH_2} + \mathrm{Br_2} + 4\mathrm{KOH} \rightarrow \mathrm{R\text{-}NH_2} + \mathrm{K_2CO_3} + 2\mathrm{KBr} + 2\mathrm{H_2O}$$

Hoffmann bromamide degradation yielding primary amine with one fewer carbon

$$\mathrm{R\text{-}NH_2} + \mathrm{CHCl_3} + 3\mathrm{KOH}\ (\text{alc.}) \xrightarrow{\Delta} \mathrm{R\text{-}NC} + 3\mathrm{KCl} + 3\mathrm{H_2O}$$

Carbylamine test for primary amines producing foul-smelling isocyanides

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Hoffmann degradation drops ONE carbon atom! E.g. Ethanamide (2 carbons) gives Methanamine (1 carbon).

• **Alert:** [Valuation Rubric] Step 1 [Reactants & Reagents]: Write primary amide + Br_2 + 4KOH .

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Explain the preparation of primary amines by: (a) Gabriel phthalimide synthesis, (b) Hoffmann bromamide degradation. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Gabriel Phthalimide Synthesis]** Phthalimide + KOH --> potassium phthalimide --> [R-X] N-alkylphthalimide --> [aq NaOH] Primary aliphatic amine ($R\text{-}NH_2$) + sodium phthalate. Pure primary amines obtained. Aromatic amines CANNOT be prepared because aryl halides do not undergo nucleophilic substitution with phthalimide anion. [2 Marks]

• **[Step 2: Hoffmann Bromamide Degradation]** Primary amide treated with Br_2 and ethanolic NaOH: $R\text{-}CONH_2 + Br_2 + 4NaOH \rightarrow R\text{-}NH_2 + Na_2CO_3 + 2NaBr + 2H_2O$. Yields primary amine with ONE LESS carbon atom than parent amide. [2 Marks]

• **Key Exam Result:** Gabriel (pure 1° aliphatic) | Hoffmann ($R\text{-}CONH_2$ --> $R\text{-}NH_2$, 1 less carbon)

2. Explain why aliphatic amines are more basic than ammonia, whereas aromatic amines (aniline) are weaker bases than ammonia. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Aliphatic Amines vs Ammonia]** In aliphatic amines, alkyl group exerts +I electron-donating inductive effect, increasing electron density on nitrogen atom and stabilizing ammonium cation, making them stronger bases than NH_3. [1.5 Marks]

• **[Step 2: Aromatic Amines vs Ammonia]** In aniline, lone pair on nitrogen is delocalized into benzene ring over 5 resonating structures (-R effect), making it less available for protonation. Anilinium ion has only 2 resonating structures (less stable). Hence aniline is much weaker base than NH_3. [1.5 Marks]

• **Key Exam Result:** Aliphatic (+I raises basicity) > NH_3 > Aromatic (-R resonance delocalization)

3. Explain the order of basicity of methyl-substituted and ethyl-substituted amines in aqueous solution. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Basicity Orders in Water]** Methyl-substituted: $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$ (2° > 1° > 3°). Ethyl-substituted: $(C_2H_5)_2NH > (C_2H_5)_3N > C_2H_5NH_2 > NH_3$ (2° > 3° > 1°). [1.5 Marks]

• **[Step 2: Three Combined Factors]** Result of subtle balance between: 1. +I inductive effect (favors 3° > 2° > 1°). 2. Hydration energy of ammonium cation (favors 1° > 2° > 3°). 3. Steric hindrance to protonation (favors 1° > 2° > 3°). In both series, 2° amine is most basic. [1.5 Marks]

• **Key Exam Result:** Methyl: 2° > 1° > 3° | Ethyl: 2° > 3° > 1°

4. Explain Carbylamine test (isocyanide test) for primary amines with chemical equation. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Reaction]** Primary aliphatic or aromatic amine heated with chloroform and alcoholic KOH: $R\text{-}NH_2 + CHCl_3 + 3KOH(alc) \rightarrow$ [Heat] R-NC (carbylamine or isocyanide) + $3KCl + 3H_2O$. $C_6H_5NH_2 + CHCl_3 + 3KOH \rightarrow C_6H_5NC$ (phenyl isocyanide) + $3KCl + 3H_2O$. [1.5 Marks]

• **[Step 2: Observation & Specificity]** Extremely foul and repulsive odor of isocyanide. 2° and 3° amines DO NOT give this test. [0.5 Mark]

• **Key Exam Result:** 1° Amine + $CHCl_3$ + 3KOH --> R-NC (extremely foul odor)

5. Explain Hinsberg's test to distinguish between primary, secondary, and tertiary amines with equations. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Hinsberg Reagent]** Benzenesulphonyl chloride ($C_6H_5SO_2Cl$). [0.5 Mark]

• **[Step 2: Differentiation]** 1° Amine: Forms N-alkylbenzenesulphonamide, which has an acidic hydrogen on nitrogen and dissolves completely in aqueous alkali (NaOH). 2° Amine: Forms N,N-dialkylbenzenesulphonamide, which lacks an acidic hydrogen and remains insoluble in alkali. 3° Amine: Has no hydrogen on nitrogen and does not react with Hinsberg reagent at all. [2.5 Marks]

• **Key Exam Result:** 1° (soluble in alkali) | 2° (insoluble in alkali) | 3° (no reaction)

6. Explain the reaction of primary, secondary, and tertiary aliphatic amines with nitrous acid (HNO2). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: With Nitrous Acid ($NaNO_2$ + HCl)]** 1° Aliphatic amine: R-NH_2 + HNO_2 --> [R-$N_2^+Cl^-$] --> R-OH + N_2 (brisk effervescence) + HCl. 1° Aromatic amine: $C_6H_5NH_2$ + HNO_2 (0-5°C) --> $C_6H_5N_2^+Cl^-$ (stable benzenediazonium chloride). 2° Amine: R_2NH + HNO_2 --> R_2N-N=O (N-nitrosamine, yellow oily liquid). 3° Amine: Forms soluble nitrite salts. [3 Marks]

• **Key Exam Result:** 1° gives N_2 gas | 1° aromatic gives diazonium salt | 2° gives yellow oil

7. Why is aniline acetylated before carrying out nitration and bromination? (Protection of -NH2 group). [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Reason for Acetylation]** The -NH_2 group in aniline is exceptionally activating due to strong +R resonance, causing poly-bromination (2,4,6-tribromoaniline) and oxidation during nitration. Acetylation with $(CH_3CO)_2O$ converts -NH_2 into -$NHCOCH_3$ (acetanilide). Lone pair on nitrogen delocalizes into carbonyl oxygen, dampening electron donation into ring. Subsequent bromination yields mono-substituted p-bromoacetanilide, which on acid hydrolysis yields pure p-bromoaniline. [2 Marks]

• **Key Exam Result:** Protects -NH_2 to prevent polysubstitution and oxidation, yielding mono-product